

Roll No

ODD END SEMESTER EXAMINATION DECEMBER- 2024

Name of the Course: B. Tech (Mechanical Engineering)
Name of the Paper: Automotive Systems

Semester: 3rd
Paper Code: TME-312

Time: 3 Hours

Maximum Marks: 100

Note:-

- (i) All questions are compulsory.
- (ii) Answer any two sub questions among a, b & c in each main question.
- (iii) Total marks in each main question are twenty.
- (iv) Each question carries 10 marks.

Q1	(20marks)	
(a)	Discuss in detail about the term 'Automobile'.	[CO1]
(b)	Discuss in detail about Automobile Industry.	
(c)	Discuss in detail about the History of Automobiles.	
Q2	(20 marks)	
(a)	Discuss in detail about the concept and functions of 'Clutch'.	[CO2]
(b)	Explain in details the functions of Steering, Electrical Systems, Cooling Systems and Fuel System of an Automobile.	
(c)	Describe in detail about the construction and working of 'Torque Converter'.	
Q3	(20 marks)	
(a)	Classify the automobile based on construction and use.	[CO3]
(b)	Describe the Architecture of automobile and factors affecting it.	
(c)	What are different subsystems of automobile?	
Q4	(20 marks)	
(a)	What are differences between conventional and alternative powertrains?	[CO4]
(b)	Explain the difference between hybrid and electric vehicle and their limitations.	
(c)	Compare the environmental impact of conventional, hybrid and electric vehicles.	
Q5	(20 marks)	
(a)	What are the different benchmarks and career opportunities in automotive industry?	[CO5] [CO6]
(b)	How the automobiles are important for society?	
(c)	Explain in detail about the standards in automobile and their role.	

